AI EXPLAINED SIMPLY

Run AI on Your Laptop, Offline: Setup Guide & Size Cheat Sheet

A quick companion to the video. Every figure below comes from the tools' own pages or the Ollama model library; the memory advice is a rule of thumb, not a guarantee.

The idea

A local model is a file. You download it once, then your own computer runs it. After the download, chatting works with no internet.

The three free tools

Tool	What it is	Offline?	Notes
Ollama	Open-source runner with a local API	Yes, after you pull a model	MIT license. Install, pull a model, chat from the terminal.
LM Studio	Desktop app with a model browser	Yes ("can operate entirely offline" once you have the model files)	Mac: Apple silicon + macOS 14 or newer. Windows: x64 needs AVX2; 16 GB RAM and 4 GB VRAM recommended.
llama.cpp	"LLM inference in C/C++", the engine under many tools	Yes	MIT license. Apple silicon, CPUs (AVX/AVX2/AVX512), Nvidia GPUs (CUDA). 1.5-bit to 8-bit quantization.

Download size cheat sheet (Ollama library)

Model	Parameters	Download

| Phi-3 | 3.8B | about 2.3 GB |

| Mistral | 7B | about 4.1 GB |

| Llama 3 | 8B | about 4.7 GB |

| Llama 3 | 70B | about 40 GB |

Rule of thumb from community guides: the file size approximates the free memory you need, plus a little for the conversation. If a model does not fit on your GPU, Ollama can offload part of it to system RAM (slower, still works).

Your 4-step plan

1. Check how much memory your laptop has.
2. Install Ollama or LM Studio.
3. Start with a small model (around 3B parameters).
4. Move up only if you want more quality and your memory allows it.

Honest limits

- Local models are smaller than the biggest cloud models; for hard reasoning and long documents the cloud is still stronger.
- Speed depends on your chip, memory and model. Test on your own machine (about ten minutes).

Sources

- Ollama: https://github.com/ollama/ollama
- llama.cpp: https://github.com/ggml-org/llama.cpp
- LM Studio system requirements: https://lmstudio.ai/docs/app/system-requirements
- Model sizes: Ollama model library (https://ollama.com/library)

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